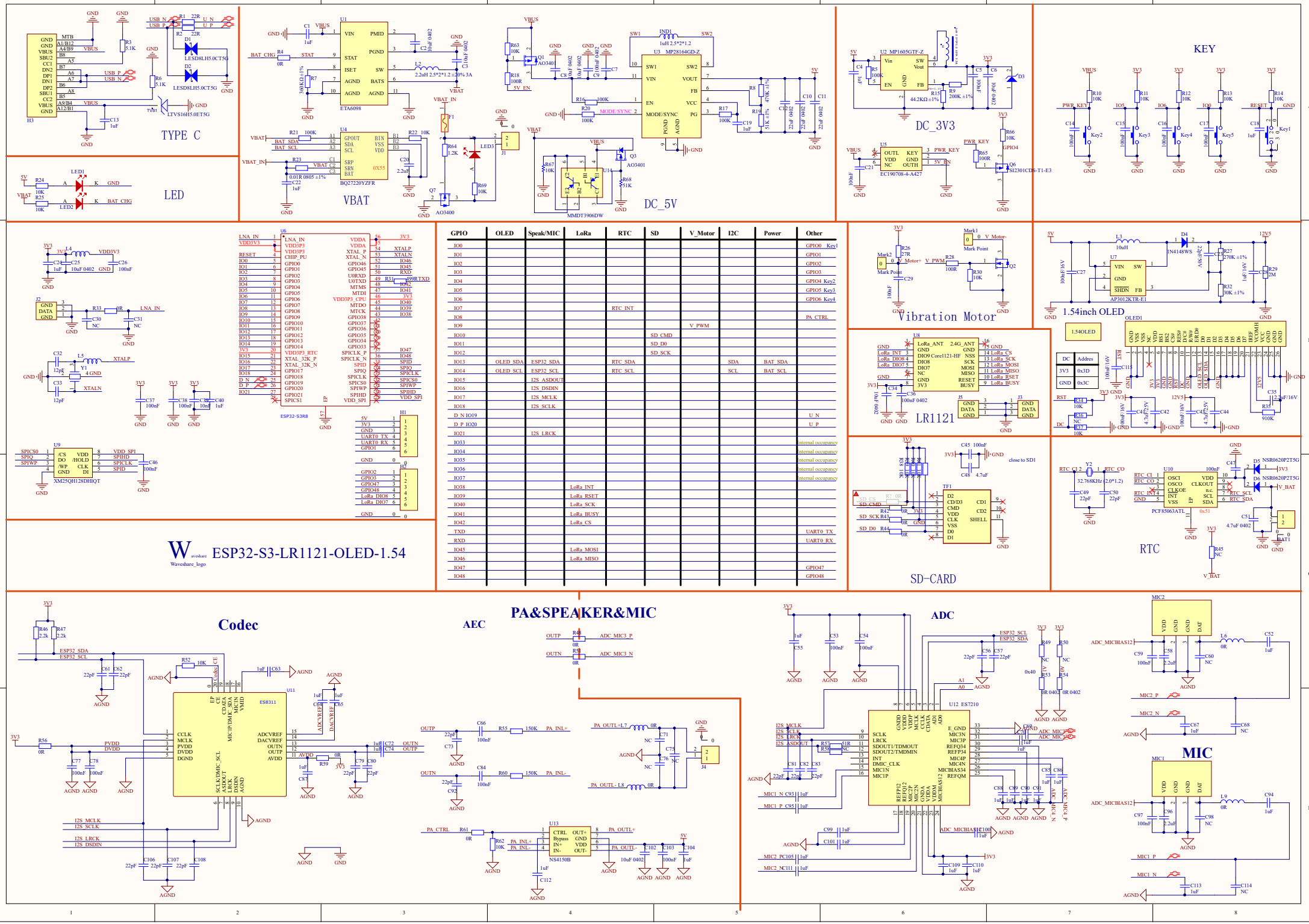
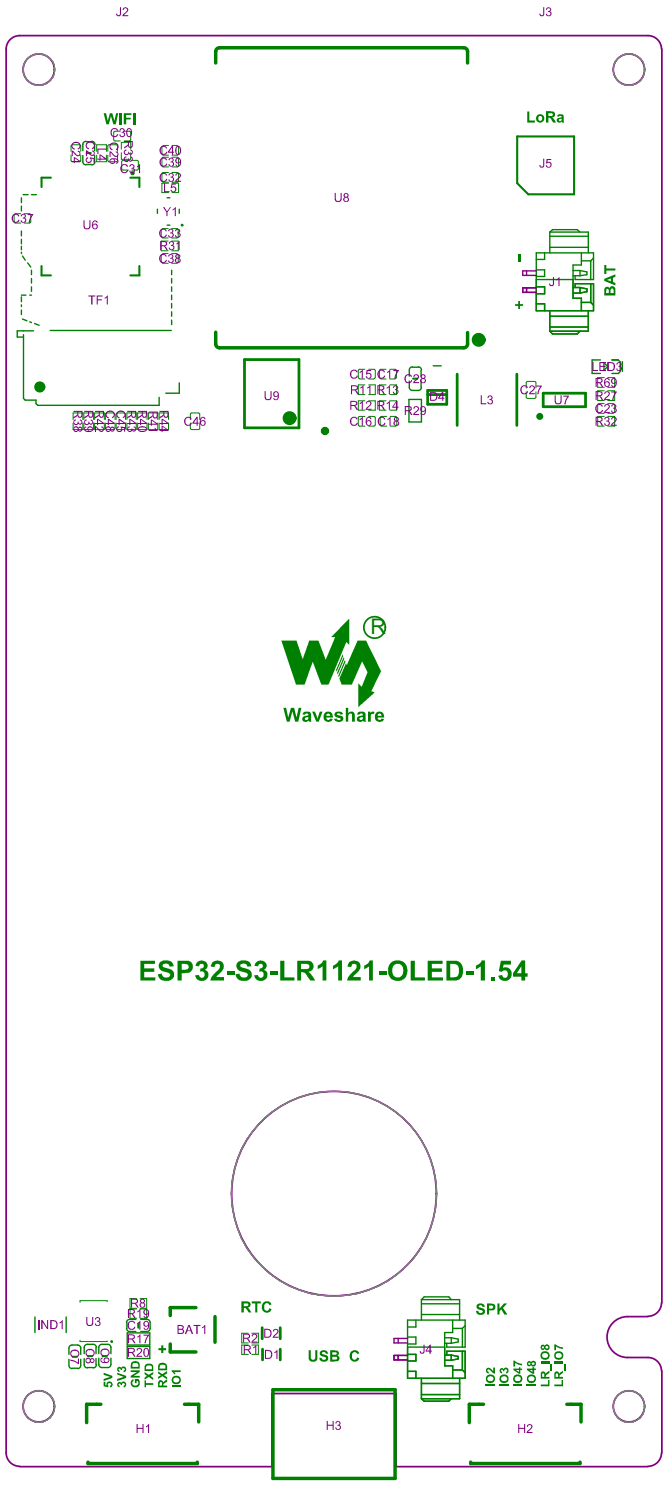
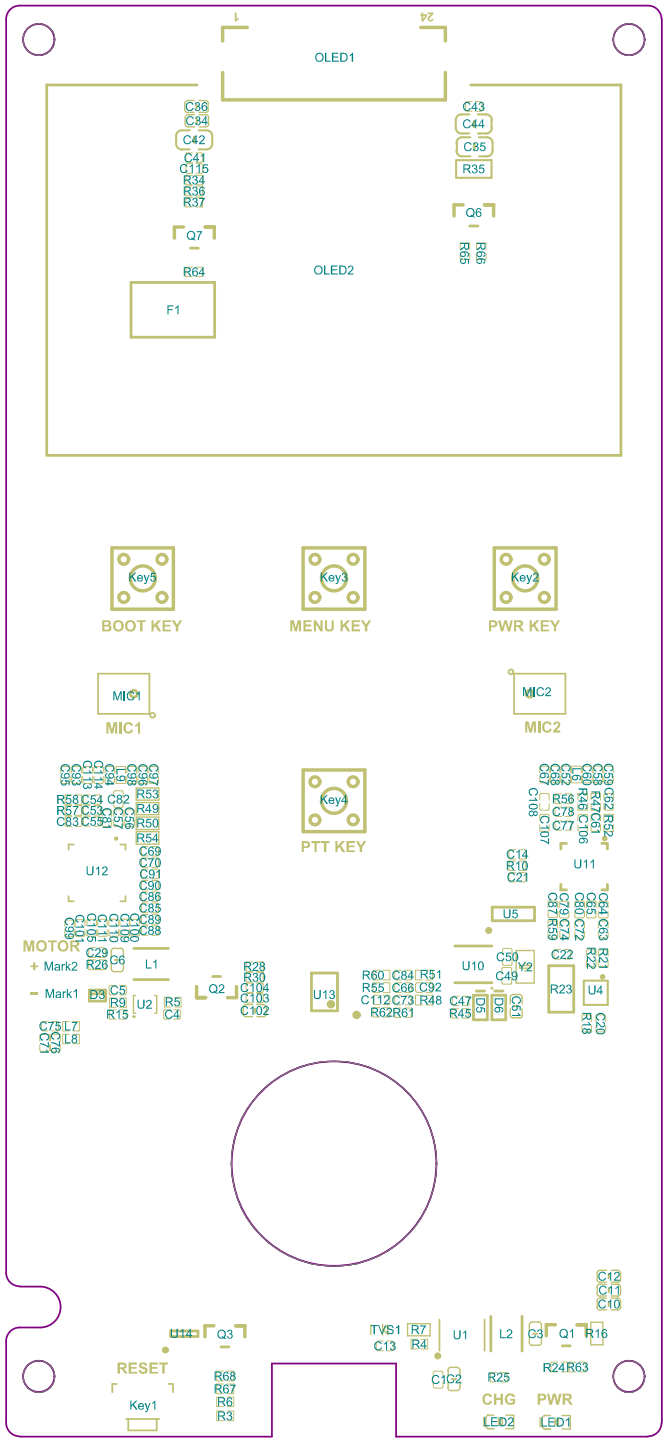

TYPE C
LED
VBAT
ETA6098
BQ27220YZFR
AO3400
AO3401
MMDT3906DW
DC_5V
DC_3V3
EC190708-4-A427
SI2301CDS-T1-E3
KEY
Vibration Motor
1.54inch OLED
AP3012KTR-E1
LR1121
SD-CARD
RTC
PCF85063ATL
ESP32-S3R8
XM25QH128DHIQT
Waveshare_logo
ESP32-S3-LR1121-OLED-1.54
PA&SPEAKER&MIC
Codec
ES8311
AEC
NS4150B
ADC
ES7210
MIC
GPIO | OLED | Speak/MIC | LoRa | RTC | SD | V_Motor | I2C | Power | Other
IO0 | GPIO0_Key1
IO1 | GPIO1
IO2 | GPIO2
IO3 | GPIO3
IO4 | GPIO4_Key2
IO5 | GPIO5_Key3
IO6 | GPIO6_Key4
IO7 | RTC_INT
IO8 | PA_CTRL
IO9 | V_PWM
IO10 | SD_CMD
IO11 | SD_D0
IO12 | SD_SCK
IO13 | OLED_SDA | ESP32_SDA | RTC_SDA | SDA | BAT_SDA
IO14 | OLED_SCL | ESP32_SCL | RTC_SCL | SCL | BAT_SCL
IO15 | I2S_ASDOUT
IO16 | I2S_DSDIN
IO17 | I2S_MCLK
IO18 | I2S_SCLK
D_N IO19 | U_N
D_P IO20 | U_P
IO21 | I2S_LRCK
IO33 | internal occupancy
IO34 | internal occupancy
IO35 | internal occupancy
IO36 | internal occupancy
IO37 | internal occupancy
IO38 | LoRa_INT
IO39 | LoRa_RSET
IO40 | LoRa_SCK
IO41 | LoRa_BUSY
IO42 | LoRa_CS
TXD | UART0_TX
RXD | UART0_RX
IO45 | LoRa_MOSI
IO46 | LoRa_MISO
IO47 | GPIO47
IO48 | GPIO48


J2
J3
WIFI
LoRa
J5
U8
U6
Y1
C37
TF1
J1
BAT
U9
L3
U7
Waveshare
ESP32-S3-LR1121-OLED-1.54
IND1
U3
BAT1
RTC
USB C
SPK
J4
5V
3V3
GND
TXD
RXD
IO1
IO2
IO3
IO47
IO48
LR_IO8
LR_IO7
H1
H3
H2


OLED1
OLED2
F1
Key5
BOOT KEY
Key3
MENU KEY
Key2
PWR KEY
MIC1
MIC2
Key4
PTT KEY
U12
U11
U5
U10
U13
U4
MOTOR
+ Mark2
− Mark1
Q2
U2
L1
Key1
RESET
Q3
U14
TVS1
U1
L2
Q1
R24R63
R25
CHG
PWR
LED2
LED1